ARSH PATEL

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SUMMARY

Computer Science & Engineering student with hands-on experience in Python backend development, RESTful API design, and Django and FastAPI services, plus ETL workflows, databases, and Dockerized applications. Strong in data structures, algorithms, and modular design; seeking a Software Engineer role building reliable, scalable backend systems.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering **CGPA: 8.67/10**
Sharad Institute of Technology College of Engineering 2023 - Expected 2027

SKILLS

Programming Languages: Python, Java, C++, SQL
Backend & Databases: Django, FastAPI, REST APIs, API Design, MySQL, MongoDB, Firebase
CS Fundamentals: Data Structures & Algorithms, OOP
DevOps & Cloud: Docker, Git, GitHub, GitHub Actions, CI/CD, Azure
AI/ML: Machine Learning, PyTorch, NLP, Transformers, LLMs, LangChain

EXPERIENCE

Machine Learning Intern Jun 2025 - Jul 2025
RacksonsIT Solutions

- Engineered a CNN traffic sign classifier reaching 98% accuracy using data preprocessing and image augmentation.
- Raised accuracy from 93% to 98% (+5 points) via data analysis, feature engineering, and hyperparameter tuning.
- Collaborated in a 4-member team using Git, code reviews, and Agile workflows; documented experiments.

PROJECTS

AI-Powered Text Summarization API Jan 2025
Python, FastAPI, Docker, MLflow, Hugging Face, T5 Transformers

- Created a FastAPI model-serving API for real-time abstractive summarization using T5 Transformers.
- Enabled iterative retraining on 14,700+ samples through a dedicated /train endpoint.
- Containerized inference with Docker, optimized for CPU-only environments to reduce GPU dependency.

Hierarchical Multi-Agent System – LLM Orchestration Oct 2025
Python, LangGraph, LangChain, Streamlit

- Architected an 8-node, 3-tier hierarchical multi-agent workflow (1 CEO orchestrator, 2 team leaders, 4 specialists) using LangGraph and LangChain for task delegation.
- Implemented a 4-stage pipeline (planning, research, writing, consolidation) with a centralized HierarchicalState tracking agent transitions and outputs.
- Developed a modular reporting platform with 2 interfaces (CLI and Streamlit UI) supporting configurable LLM providers and real-time agent-message tracking.

Dockerfile Explainer – Generative AI Dec 2025
Python, LLaMA, LoRA, PEFT, 4-bit Quantization

- Fine-tuned a 3.5B parameter LLaMA model with LoRA and PEFT, training 8.39 million parameters.
- Reduced peak VRAM from 12.57 GB to 3.96 GB (68.5% lower) using 4-bit quantization.
- Generated multi-stage Dockerfile explanations at 19.33 seconds average inference latency.

ACHIEVEMENTS

- **PATH – Public Transportation Assistance Hub | Semi-Finalist, e-Yantra (IIT Bombay)**, 2024 - 2025. Designed a rural public transportation accessibility solution under IIT Bombay mentorship; reached the semi-finals among 400 teams across India.
- **CelestiQ | Co-Author, Patent**, Mar 2025. Developed the Django-based backend for the Aptitude Assessment Module and structured the Group Discussion (GD) framework for candidate interaction and evaluation. Co-authored 1 published design patent for an AI-driven interview evaluation and recruitment assistance system.